

	1	10	20	30	40	50
Corynebacterium glutamicum_groL	.MAKL	TAFDQD	AREGILR	GVDALANA	VKVTLGPRGRNVVLDKAF	GGFLVTNDGVTIARDI
Bifidobacterium animalis_groL	.MAKI	IEYDEE	ARQGMLA	GLDKLADT	VKVTLGPKGRNVVLDKTY	GAPTITNDGVSIAKEI
Escherichia coli_groL	MAAKD	VKFGND	ARVKMLR	GVNVLADA	VKVTLGPKGRNVVLDKSF	GAPTITKDGVSVAREI
Thermus thermophilus_groL	.MAKI	LVFDEA	ARRALER	GVNAVANA	VKVTLGPRGRNVVLEKKG	FSPTITKDGVTVAKEV

	60	70	80	90	100	110
Corynebacterium glutamicum_groL	DLEDPF	ENLGAQLVKS	SAVKTNDI	AGDGTTTATLLAQA	LIIEGLRNVA	AGANPMELNKGI
Bifidobacterium animalis_groL	DLEDPF	ERIGAEIVKE	VAKRRTDD	VAGDGTTTATVLAQ	SLVHEGLKNV	VAGSNPTALRRGI
Escherichia coli_groL	ELEDKF	ENMGAQMVKE	VASKAND	AGDGTTTATVLAQA	IIIEGLKAVA	AGMNPMDLKRGI
Thermus thermophilus_groL	ELEDH	ENIGAQLLKE	VASKTND	VAGDGTTTATVLAQA	IVREGLKNV	AGANPMLALKRGI

	120	130	140	150	160	170
Corynebacterium glutamicum_groL	SAAAEK	TEELKAR	ATEVSDTR	ETANVATVSSR	.DEVVGEIVAA	AMEKVGKDG
Bifidobacterium animalis_groL	EKASD	ALVKQLV	ASAKPVE	TKETIAATA	TISAG.DPE	VGEKIAEALD
Escherichia coli_groL	DKAVT	AAVEELK	ALSVP	PCSDSKA	IAQVGTISAN	SDETV
Thermus thermophilus_groL	EKA	VEAAVEK	IKALA	IPVEDRKA	IEEVA	TISAN.DPE

	180	190	200	210	220	230
Corynebacterium glutamicum_groL	QSIETA	LEVT	EGISF	DKGYLS	PFYINDN	DTQQAVLDN
Bifidobacterium animalis_groL	NRFG	LDLDF	TEGMR	FDKGYI	SPYFVTNA	EDQTAVLDD
Escherichia coli_groL	TGLQ	DELDV	VEGMD	FDKGYLS	PFYINKPE	ETGAVELES
Thermus thermophilus_groL	KSLE	TELK	FVEGY	QFDKGYI	SPYFVTNP	ETMEAVLEDA

	240	250	260	270	280	290
Corynebacterium glutamicum_groL	VESNR	PLLI	IAEDVE	GEPL	QTLV	VNSIRK
Bifidobacterium animalis_groL	MKTG	KPLLI	IAEDVD	GEALP	TLIL	NNIRG
Escherichia coli_groL	AKAG	KPLLI	IAEDVE	GEALAT	TVVNT	MRGIV
Thermus thermophilus_groL	AQTG	KPLLI	IAEDVE	GEALAT	TVVNT	KLRL

	300	310	320	330	340	350
Corynebacterium glutamicum_groL	VDP	EVGIN	LNEAGE	EVFGT	ARRITV	SKDETH
Bifidobacterium animalis_groL	VSED	LGKLD	SIDL	SVFGT	AKKVI	VS
Escherichia coli_groL	ISEE	IGME	LEKAT	LEDLG	AKRV	VINKD
Thermus thermophilus_groL	ISEE	LGFK	LENAT	LMSLGR	AERV	IRITK

	360	370	380	390	400	410
Corynebacterium glutamicum_groL	DREKA	ERLAKL	SGCT	AVIRVGA	ATEETE	VNDRKL
Bifidobacterium animalis_groL	DREKL	QERLAK	LGGV	AVIKVGA	ATEVEA	KERKHRI
Escherichia coli_groL	DREKL	QERVAK	LGGV	AVIKVGA	ATEVEA	KERKHRI
Thermus thermophilus_groL	AREKL	QERLAK	LGGV	AVIRVGA	ATEETEL	KEKKHRI

	420	430	440	450	460	470
Corynebacterium glutamicum_groL	QIAET	LKAYAE	EEFG	DQKV	GVRAL	TALGKE
Bifidobacterium animalis_groL	QAAEK	VEKDFN	.LEGG	DEAT	GAAIV	FSGIEA
Escherichia coli_groL	RVASK	LADLRG	.QNE	DQNV	GIVK	VALRAMEA
Thermus thermophilus_groL	RAIS	AVEELIK	LEGG	DEAT	GAKIV	RRALEEP

	480	490	500	510	520	530
Corynebacterium glutamicum_groL	FNAAT	LEYGN	LINDG	VIDP	VKTHSA	VNATSV
Bifidobacterium animalis_groL	FNAAT	DTYED	LMAAG	VTDP	VKTR	SALQNA
Escherichia coli_groL	YNAAT	EEYGN	MIDMG	ILDPT	KVTR	SALQYA
Thermus thermophilus_groL	FNAAT	GEEF	VDMVEA	GIVDP	AKVTR	SALQNA

Corynebacterium glutamicum_groL	HH.....
Bifidobacterium animalis_groL	.QDMGY....
Escherichia coli_groL	MGGMGGMGGMM
Thermus thermophilus_groL	AGDMDF.....